Lattice Sigma Terms as an Anchor for the Dense Nuclear Matter Equation of State

Oleg Kirichenko

Independent Researcher; urevich55@gmail.com

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Abstract

A phenomenological Vacuum Mass Fraction (VMF) model is proposed, in which the hadron mass is separated into two components: (1) the current mass, determined by the explicit breaking of chiral symmetry via sigma terms, and (2) the residual nonperturbative component M_Ω, generated by confinement, gluon field energy, and the QCD trace anomaly.

Using lattice QCD data for the pion-nucleon ($\sigma_{\pi N} \approx 44$ MeV) and strange ($\sigma_{sN} \approx 30$ MeV) sigma terms, the vacuum value of the nonperturbative component is fixed at $M_{\Omega,0} = 859 \pm 8$ MeV, constituting $\sim 91\%$ of the nucleon mass.

A series of EOS prototypes for dense nuclear matter is constructed with a two-parameter ansatz for the in-medium modification $M_\Omega(n_B)$. It is shown that: (i) minimal realizations with constant vector repulsion yield a superluminal speed of sound; (ii) density-dependent saturation of the vector field resolves the causality problem ($c_s^2 \leq 1$); (iii) the problem of an overly stiff equation of state is resolved by introducing a first-order phase transition to the conformal QGP limit, reducing the maximum neutron star mass to a realistic $M_{\max} \approx 2.3\, M_\odot$.

The model is strictly bounded by input data: the vacuum scale $M_{\Omega,0}$ is fixed by lattice calculations and is not a free fitting parameter.

Keywords: hadron mass, sigma terms, equation of state, neutron stars, nonperturbative QCD

Contents

1 Introduction

1.1 The Nucleon Mass Problem

The proton mass $M_N \approx 938.3$ MeV is only to a small extent determined by the current quark masses. According to the Ji decomposition [1], QCD sum rules [2], and the analysis of the energy-momentum tensor trace, up to $\sim 90\%$ of the mass is generated by nonperturbative mechanisms: gluon field energy and the trace anomaly.

At the operator level:

$$T^{\mu}{}_{\mu} = \frac{\beta(g)}{2g}\, G^a_{\alpha\beta} G^{a\alpha\beta} + (1+\gamma_m(g)) \sum_q m_q \bar{q} q\,. \tag{1}$$

Even in the chiral limit ($m_q \to 0$), the gluon term remains—mass is generated nonperturbatively.

1.2 Aim of the Work

We propose using standard QCD sigma terms to define a dimensionless parameter f_Ω, which characterizes the fraction of the hadron mass not reducible to current quark masses, and to investigate the behavior of this parameter in a dense baryon medium through its impact on the equation of state of nuclear matter.

2 Mass Decomposition via Sigma Terms

2.1 Definitions

Via the Feynman–Hellmann theorem, the standard sigma terms are:

$$\sigma_{qH} = m_q \frac{\partial M_H}{\partial m_q}\,. \tag{2}$$

The residual nonperturbative mass of hadron H:

$$\boxed{M_\Omega^{(H)}[\mathcal{F}] \equiv M_H - \sum_{q\in\mathcal{F}} \sigma_{qH}} \tag{3}$$

The dimensionless fraction:

$$\boxed{f_\Omega^{(H)}[\mathcal{F}] = 1 - \frac{1}{M_H} \sum_{q\in\mathcal{F}} \sigma_{qH}} \tag{4}$$

2.2 Evaluation for the Nucleon

Using lattice data [4, 5]:

Quantity	Value	Source
$\sigma_{\pi N}$	44 ± 3 MeV	Gupta et al. (2021) [5]
σ_{sN}	30 ± 7 MeV	Agadjanov et al. (2023) [4]
σ_{heavy}	~ 6 MeV	PDG matching [6]
$\sum_q \sigma_{qN}$	80 ± 15 MeV	Sum

Table 1: Sigma-term input data for the nucleon.

We obtain:

$$f_\Omega^{(N)} \approx 1 - \frac{80 \pm 15}{939} \approx 0.91 \pm 0.02\,, \qquad M_{\Omega,0} = 859 \pm 8 \text{ MeV}\,. \tag{5}$$

Within this phenomenological decomposition, about 90% of the nucleon mass is associated with nonperturbative QCD contributions not reducible to explicit current quark masses.

2.3 Connection to the Ji Decomposition

In the standard decomposition:

$$M_H = M_H^{\text{q,kin}} + M_H^{\text{g}} + M_H^{\text{q,mass}} + M_H^{\text{anom}}\,, \tag{6}$$

our residual mass corresponds to:

$$M_\Omega^{(H)} \sim M_H^{\text{g}} + M_H^{\text{anom}} + M_H^{\text{conf}}\,. \tag{7}$$

This is a phenomenological identification, not a theorem (cf. bag models [3]). Its value lies in organization: a single variable f_Ω compresses the question of "what fraction of mass does not reduce to current masses" into a single number.

2.4 Status of the Statement

The quantity $f_\Omega^{(N)}$ is *not* a new fundamental operator or conserved charge. It is an EFT parameter that organizes the hadron mass budget. Its value is stable upon fixing the flavor set $\mathcal{F}$ and is compatible with independent lattice decompositions.

3 Field Structure of the Model

The action of the VMF model has the standard form of GR + a canonical scalar field:

$$S = \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}(R - 2\Lambda_{\text{eff}}) - \frac{1}{2}\partial_\mu\mathcal{W}\partial^\mu\mathcal{W} - V(\mathcal{W}) + \mathcal{L}_m\right]. \tag{8}$$

The Einstein tensor is unmodified, and the causal cone is preserved. The construction is mathematically equivalent to standard scalar-tensor models with conformal coupling $\omega \to \infty$ (the Brans–Dicke limit [10]), guaranteeing $\gamma_{\text{PPN}} = 1$ and automatic compatibility with Solar System constraints (Cassini: $|\gamma - 1| < 2.3 \times 10^{-5}$ [7]).

The vacuum state is defined as $\mathcal{W}_0$, where $V'(\mathcal{W}_0) = 0$, $V''(\mathcal{W}_0) > 0$. As $n_B \to 0$, all medium fields vanish, and the model is identical to classical GR.

4 Equation of State of Dense Matter

4.1 Scalar Field as an Order Parameter

We postulate that the scalar field $\mathcal{W}$ acts as a macroscopic order parameter for the nonperturbative QCD condensate:

$$M_\Omega = g_\Omega \langle \mathcal{W} \rangle \, . \tag{9}$$

In a dense baryon medium, the dynamics of the order parameter is determined by minimizing the effective potential:

$$V_{\text{eff}}(\mathcal{W}, n_B) = V(\mathcal{W}) + n_B \, M^*(\mathcal{W}) \, . \tag{10}$$

Since the exact form of $V(\mathcal{W})$ from first principles is unknown, we parameterize the *solution* through a phenomenological ansatz:

$$M_\Omega(n_B) = M_{\Omega,0} \left(1 + \kappa_2 \frac{n_B}{n_0} \right)^{-\kappa_1/\kappa_2} , \tag{11}$$

where κ_1, κ_2 are phenomenological medium-response parameters, *not derived* from first principles of QCD.

The current component depends on density through the chiral condensate:

$$M_{\text{current}}(n_B) = M_{\text{current},0} \cdot \frac{\langle \bar{q}q \rangle_{n_B}}{\langle \bar{q}q \rangle_0} \, . \tag{12}$$

In the low-density approximation:

$$\frac{\langle \bar{q}q \rangle_{n_B}}{\langle \bar{q}q \rangle_0} \approx 1 - \frac{\sigma_{\pi N} \, n_B}{f_\pi^2 \, m_\pi^2} \, , \tag{13}$$

so the total effective nucleon mass takes the form:

$$M^*(n_B) \approx M_{\text{current},0} \left(1 - \frac{\sigma_{\pi N} \, n_B \, (\hbar c)^3}{f_\pi^2 \, m_\pi^2} \right) + M_\Omega(n_B) \, . \tag{14}$$

4.2 Energy Density

$$\varepsilon(n_B) = \sum_{i=n,p,e,\mu} \varepsilon_{\text{kin}}^i + V(\langle \mathcal{W} \rangle) + \varepsilon_{\text{vec}}(n_B) + \varepsilon_\rho(n_B) \, , \tag{15}$$

with density-dependent vector saturation:

$$\varepsilon_{\text{vec}}(n_B) = \frac{C_{\omega 0}}{2} \cdot g(n_B) \cdot n_B^2, \quad g(x) = \frac{1}{1 + \alpha_v (x-1)^{\nu_v}} \text{ for } x > 1 \, . \tag{16}$$

4.3 Calibration

The model is calibrated by nuclear physics, *not* pulsars:

Constraint	Value	What it fixes
Binding energy	$E/A = -16$ MeV at n_0	$C_{\omega 0}$
Pressure	$P(n_0) = 0$	Saturation
Symmetry energy	$S(n_0) = 32$ MeV	C_ρ
Vacuum mass	$M_{\Omega,0} = 859 \pm 8$ MeV	**Fixed by lattice**

Table 2: Nuclear physics calibration constraints.

4.4 The Double Counting Problem

When merging the nonperturbative QCD condensate with macroscopic EOS, a risk of double counting arises. This is resolved by the fact that $M_{\Omega,0}$ acts exclusively as a **vacuum anchor**. In vacuum ($n_B = 0$), macroscopic medium fields are absent, and $M_{\Omega,0}$ strictly corresponds to the lattice value. The interaction constants of the macroscopic sector are **renormalized** around this rigid vacuum anchor during calibration (Sec. 4). The total energy density is strictly tied to nuclear observables, eliminating physical energy duplication.

4.5 Sensitivity Analysis

The total uncertainty from first principles:

$$\Delta M_{\Omega,0} = \sqrt{(\Delta\sigma_{\pi N})^2 + (\Delta\sigma_{sN})^2} \approx \sqrt{3^2 + 7^2} \approx 7.6 \text{ MeV}\,, \tag{17}$$

which is less than 1% of the baseline $M_{\Omega,0} = 859$ MeV. The maximum neutron star mass $M_{\max}$ varies within $\pm 0.05\, M_\odot$ under extreme sigma-term shifts (1σ bounds), proving the robustness of the EFT predictions.

4.6 Model Constraints

1. Lattice QCD establishes $M_{\Omega,0} \in [851, 867]$ MeV.
2. Causality: $c_s^2 \leq 1$ at all densities.
3. Astrophysical compatibility: $M_{\max} \geq 2.01\, M_\odot$, $R_{1.4} \in [11.5, 13.0]$ km.

5 Computational Results

5.1 Best Fit Point

With the saturating vector field sector:

Parameter	Value
κ_1	0.25
κ_2	0.80
α_v	4.0
ν_v	2.0
$M_D(n_0)/M_N$	0.675
$c^2_{s,\max}$	0.393

Table 3: Best-fit parameter set.

5.2 Resolving the Overly Stiff EOS

The minimal model yielded $M_{\max} \sim 4.3$–$4.7\, M_\odot$, far exceeding the observational constraint $M_{\max} \lesssim 2.3\, M_\odot$. Since the fixed QCD input does not allow arbitrary parameter fitting, the model itself pointed to the necessity of a **first-order phase transition** to conformal QGP ($P = \varepsilon/3$) at $n_B \approx 2.0\, n_0$. Solving the TOV equations yields $M_{\max} \approx 2.3\, M_\odot$ and $R_{1.4} \approx 12$ km.

6 Extrapolation to the Black Hole Interior: A Heuristic Analysis

This section is exploratory and lies outside the strict EFT framework of Sections 2–5. All results are explicitly marked as heuristic.

6.1 Concept

One speculative hypothesis is the extrapolation of $M_\Omega(n_B) \to 0$ to its logical limit inside the event horizon. Outside the horizon ($n_B \to 0$), the model is identically GR (Schwarzschild/Kerr). Inside, at $n_B \gg n_0$, the mass melts and matter transitions to conformal QGP:

$$P = \frac{1}{3}\varepsilon, \qquad c_s^2 = \frac{1}{3} < 1 \quad \text{(causal)}\,. \tag{18}$$

6.2 Phenomenological Stitching

The VMF model alone predicts $P = \varepsilon/3$, which satisfies the strong energy condition and *inevitably leads to a singularity.* The transition to a quantum core ($P_r = -\varepsilon_{\max}$) requires physics beyond this EFT. We present a phenomenological stitching with Markov's limiting density hypothesis. This strategy is widely applied in the literature [11, 12, 13, 14]. Our contribution is the fixation of r_0 via QCD parameters rather than free fitting.

The heuristic density profile $\varepsilon(r) = \varepsilon_{\max} r_0^6/(r^3 + r_0^3)^2$ yields the Hayward metric:

$$ds^2 = -\left(1 - \frac{2GMr^2}{r^3 + r_0^3}\right) dt^2 + \left(1 - \frac{2GMr^2}{r^3 + r_0^3}\right)^{-1} dr^2 + r^2 d\Omega^2\,. \tag{19}$$

As $r \to 0$, a regular de Sitter core forms. This illustrates *mathematical compatibility* of the QCD scale with regular metrics, not a dynamical prediction.

6.3 Falsifiability

1. If NICA/FAIR do not detect in-medium mass modifications at $n_B \sim 3$–$5\,n_0$, the model is falsified.
2. Gravitational-wave echoes from a regular core in post-merger signals would support the model.
3. The exterior geometry is identically Schwarzschild/Kerr; any deviations falsify the extrapolation.

7 Observational Constraints

Gravity: LIGO-Virgo-KAGRA (O4a): 42 BH mergers, no deviations from GR [9]. Tensor mode speed: $|c_T/c - 1| \lesssim 10^{-15}$ [8]. Cassini: $|\gamma_{\rm PPN} - 1| < 2.3 \times 10^{-5}$ [7].

Hadron physics: Lattice QCD (sigma terms, gluon fractions); near-threshold J/ψ photoproduction (scalar gluon structure); neutron stars via NICER and GW ($\kappa_\Omega(n_B)$—vacuum mass response to density).

8 Connection with Existing Dense Matter Models

Similarities: Density-dependent effective mass and vector saturation are standard RMF elements [15, 16]. In-medium condensate modification relates to Brown–Rho scaling [17]. The conformal crossover is characteristic of quarkyonic matter and NJL models; see reviews [18, 19].

Key novelty: Unlike RMF models where the nucleon mass is a fitting parameter, here $M_{\Omega,0}$ is strictly fixed by lattice QCD via sigma terms. The single organizing parameter f_Ω links nuclear physics (symmetry energy) with hadron phenomenology (vector meson mass shifts).

9 Numerical Implementation and Phenomenology

Hadron	Total Mass (MeV)	M_Ω (MeV)	f_Ω (%)
Nucleon (N)	939.0	859.0	91.5
ρ-meson	775.3	695.3	89.7
ω-meson	782.6	702.6	89.8
Pion (π)	139.6	72.6	52.0
Kaon (K)	493.7	131.7	26.7

Table 4: Nonperturbative mass fractions for selected hadrons. Goldstone bosons (π, K) exhibit low f_Ω, while ordinary hadrons cluster at $\sim 90\%$.

n_B/n_0	m_ρ (MeV)	m_ω (MeV)	Mass drop (%)
0.0	775.3	782.6	0.0
1.0	658.6	664.7	-15.0
2.0	595.8	601.2	-23.2
3.0	554.3	559.3	-28.5

Table 5: In-medium vector meson mass shift: a $\sim 24\%$ drop of ρ-meson mass at $2n_0$ provides a direct signature for FAIR/HADES dielectron spectrometers.

10 Conclusion

A phenomenological model is proposed wherein the rigidly fixed value of the nonperturbative nucleon mass component, derived from lattice QCD ($M_{\Omega,0} = 859 \pm 8$ MeV), serves as a fundamental constraint on the dense matter equation of state. The concept of an in-medium modified vacuum mass $M_\Omega(n_B)$ provides a low-parameter and strictly falsifiable method to integrate the QCD energy budget into neutron star phenomenology.

The overly stiff EOS problem is alleviated by a phase transition, and causality is ensured by a density-dependent saturating vector field. The model offers concrete observational signatures for gravitational interferometers (LIGO/Virgo) and heavy-ion physics (FAIR/HADES).

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